

Lectures on Multiphysics Modeling

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Linear Elasticity 1 & 2

13-14 / 4 / 26

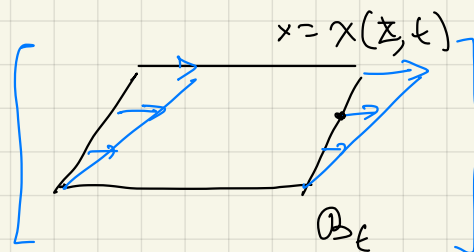
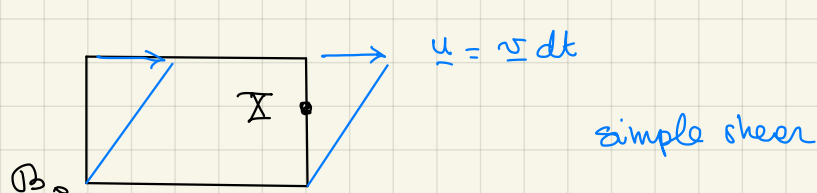
control session on 17/4/26

Linear Elasticity

Start a flow from reference configuration, and follow it for time $= dt$

Generate infinitesimal displacements $\underline{u} = \underline{v} dt$

In the reference configuration $\underline{x} = \underline{X}$, no need to distinguish between the two



The skew symmetric part of $\nabla \underline{u}$ is related to rigid rotations

The symmetric part of $\nabla \underline{u}$ is called (infinitesimal or linear) strain

and is responsible for build-up of stress $\underline{T}$

Alternatively, linear elasticity can be derived from nonlinear elasticity (small vs large deformations, see later).

• kinematics

strains $\underline{E} = \text{sym } \nabla \underline{u}$, or $E_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$

• stresses and balance (equilibrium)

$$\text{div } \underline{T} + \underline{b}_0 = 0$$

$$0 = \int_{\partial B_0} \underline{T} \underline{n} \, dA + \int_{B_0} \underline{b} \, dV$$

• Constitutive equations

$$\underline{T} = 2\mu \underline{E} - p \underline{I} \quad ; \quad \text{div } \underline{u} = 0 \quad (\text{isotropic, incompressible})$$

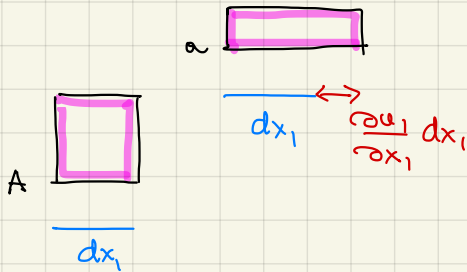
$$\underline{T} = 2\mu \underline{E} + \lambda (\text{tr } \underline{E}) \underline{I} \quad (\text{isotropic})$$

$$\underline{T} = \underline{C} \underline{E}$$

(general, $\underline{C}$ 4th-order tensor of elastic moduli)

Strains

$$E_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad \text{symmetric in } i, j$$



$$E_{11} = \frac{\partial u_1}{\partial x_1} = \frac{l - l_0}{l} = \epsilon_x$$

$$E_{12} = \frac{1}{2} \left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right) = E_{xy} = \frac{1}{2} \gamma_{xy}$$

$$\text{tr } E = \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3} = \text{"div } \underline{v} \text{ dt"}$$

$$[E] = \begin{bmatrix} \epsilon_x & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{xy} & \epsilon_y & \epsilon_{yz} \\ \epsilon_{xz} & \epsilon_{yz} & \epsilon_z \end{bmatrix}$$

Symmetry $\Rightarrow$ 6 independent components, can be represented as a column vector

$$[E] = \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \epsilon_{yz} \\ \epsilon_{xz} \\ \epsilon_{xy} \end{bmatrix}$$

(Voigt notation)

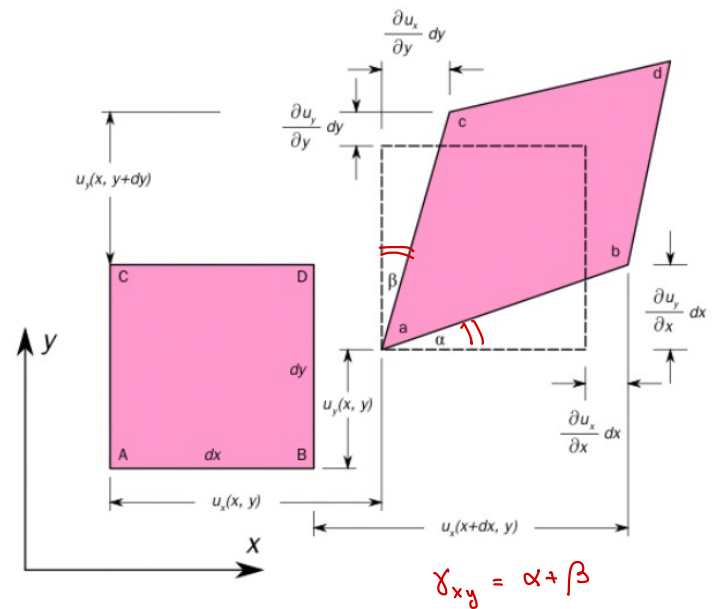


Figure: Geometric meaning of strain components (from Wikipedia)

γ_{xy} angle reduction between two orthogonal fibers

volume change per unit ref. volume

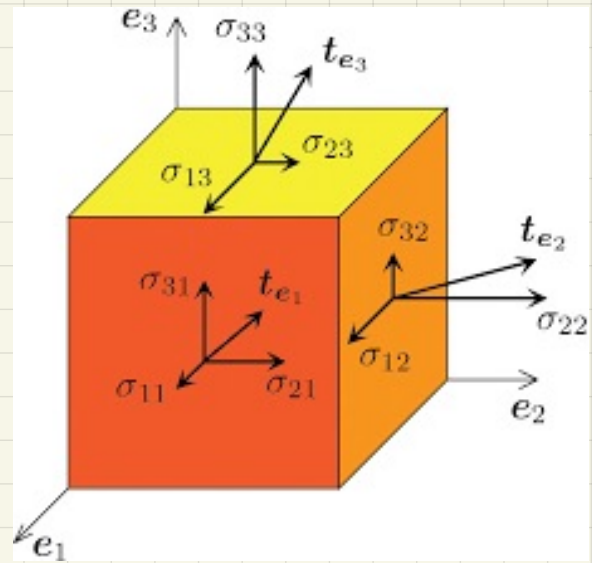
$(\text{tr } E) dV_0$ infinitesimal volume change of elementary volume dV_0

Stresses

$$\underline{t}_n = \underline{T} \underline{n}$$

$$T_{ij} = \underline{T} \underline{e}_j \cdot \underline{n} = T_{ji} \quad \text{symmetric}$$

$$[T] = \begin{bmatrix} \sigma_x & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_y & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_z \end{bmatrix}$$



Symmetry $\Rightarrow$ 6 independent components, can be represented as a column vector

$$[\sigma] = \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \sigma_{yz} \\ \sigma_{xz} \\ \sigma_{xy} \end{bmatrix} \quad \begin{matrix} \tau_{yz} \\ \tau_{xz} \\ \tau_{xy} \end{matrix} \quad (\text{Voigt notation})$$

$$\underline{T} = \text{sph } \underline{T} + \text{dev } \underline{T}$$

$$\text{sph } \underline{T} = \frac{1}{3} (\text{tr } \underline{T}) \underline{I}$$

spherical or hydrostatic part of $\underline{T}$

$$\text{dev } \underline{T} = \underline{T} - \frac{1}{3} (\text{tr } \underline{T}) \underline{I}$$

deviatoric or traceless part of $\underline{T}$

$$\sigma_{vm} = \sqrt{\frac{3}{2}} |\text{dev } \underline{T}| = \sqrt{\frac{3}{2}} (\text{dev } \underline{T} \cdot \text{dev } \underline{T})^{1/2}$$

(constant $\sqrt{3}/2$ so that, for uniaxial tension, $\sigma_{vm} = \sigma$)

$$\underline{\text{EX}} \quad \underline{T} = \begin{bmatrix} \sigma & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \frac{1}{3} (\text{tr } \underline{T}) \underline{I} + \left(\underline{T} - \frac{1}{3} (\text{tr } \underline{T}) \underline{I} \right) = \begin{bmatrix} \sigma/3 & 0 & 0 \\ 0 & \sigma/3 & 0 \\ 0 & 0 & \sigma/3 \end{bmatrix} + \underbrace{\begin{bmatrix} \frac{2}{3}\sigma & 0 & 0 \\ 0 & -\sigma/3 & 0 \\ 0 & 0 & -\sigma/3 \end{bmatrix}}_{\text{dev } \underline{T}}$$

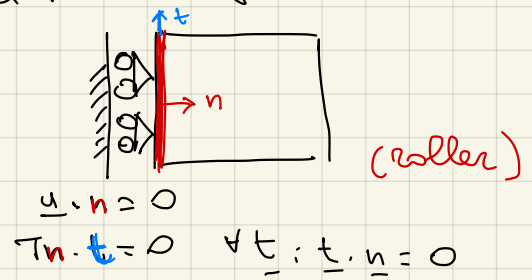
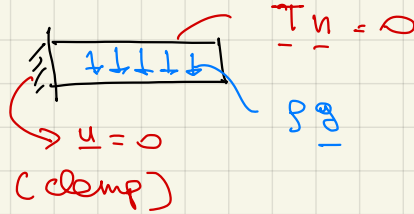
$$\text{dev } \underline{T} \cdot \text{dev } \underline{T} = \frac{4}{9} \sigma^2 + \frac{\sigma^2}{9} + \frac{\sigma^2}{9} = \frac{6}{9} \sigma^2 = \frac{2}{3} \sigma^2 \Rightarrow |\text{dev } \underline{T}| = \sqrt{\frac{2}{3}} \sigma \Rightarrow \sigma_{vm} = \sigma$$

Equilibrium

$$dW \underline{I} + b_0 = 0 \quad b_0 \text{ free p. unit reference volume}$$

+ boundary conditions on displacement and tractions

E.g., mixed problem



Constitutive equations : linearly elastic material (Hooke's law)

Stress proportional to strain

For isotropic materials, two independent parameters

$$\underline{T} = 2\mu \underline{E} + \lambda (\text{tr} \underline{E}) \underline{I} \quad \lambda, \mu \text{ Lamé constants } (\mu = G \text{ shear modulus})$$

Alternative representations for the isotropic constitutive equation (Voigt)

$$\begin{matrix} [\underline{\sigma}] & = & [\underline{C}] [\underline{\epsilon}] \\ 6 \times 1 & & 6 \times 6 \quad 6 \times 1 \end{matrix}$$

$$[\underline{C}] = \frac{E_Y}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & 1-2\nu & 0 & 0 \\ 0 & 0 & 0 & 0 & 1-2\nu & 0 \\ 0 & 0 & 0 & 0 & 0 & 1-2\nu \end{bmatrix}$$

$$E_Y = \frac{\mu(2\mu + 3\lambda)}{\mu + \lambda} \quad \text{Young modulus}$$

$$\nu = \frac{\lambda}{2(\mu + \lambda)} \quad \text{Poisson ratio}$$

$$\underline{T} = 2\mu \underline{E} + \lambda (\text{tr} \underline{E}) \underline{I}$$

To discuss the physical meaning of Young's modulus and Poisson's ratio we need to invert this:

$$\underline{T} = \underline{C} \underline{E}$$

$\underline{C}$ tensor of elastic moduli (stiffnesses)

$$\underline{E} = \underline{C}^{-1} \underline{T}$$

$\underline{C}^{-1}$ — — — compliances (deformability)

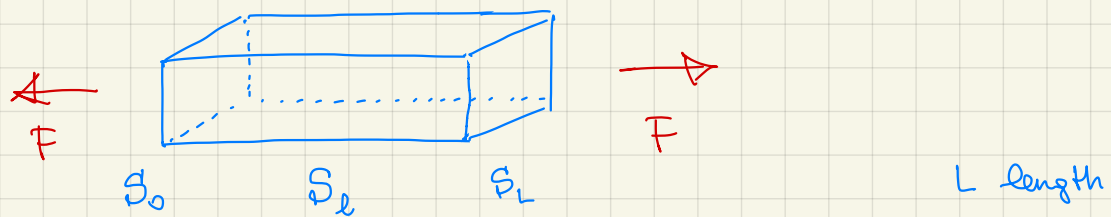
Find inverse relation $\underline{E}$ in terms of $\underline{T}$:

$$\underline{E} = \frac{1}{2\mu} \left[\underline{T} - \frac{\lambda}{2\mu + 3\lambda} (\text{tr} \underline{T}) \underline{I} \right] \quad (*)$$

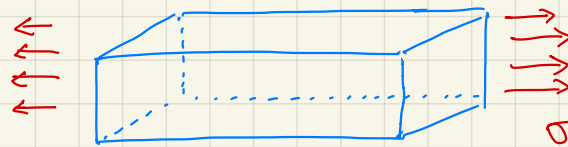
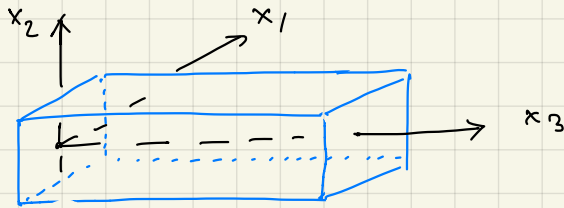
Observe that $\text{tr} \underline{T} = 2\mu \text{tr} \underline{E} + \lambda (\text{tr} \underline{E}) (\text{tr} \underline{I}) = (2\mu + 3\lambda) \text{tr} \underline{E}$

$$\text{then } \frac{1}{2\mu} \left(\underline{T} - \frac{\lambda}{2\mu + 3\lambda} (\text{tr} \underline{T}) \underline{I} \right) = \underline{E} \quad \checkmark$$

Some simple solutions: uniaxial extension, no body forces.



Uniaxial extension



$\underline{T}(\pm \underline{e}_3) = \pm \sigma \underline{e}_3$ on the bases S_0, S_L
 $\underline{T}\underline{n} = 0$ on lateral surface S_l

$$\underline{T} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sigma \end{bmatrix} \text{ satisfies equilibrium } \operatorname{div} \underline{T} = 0 \text{ and BC's}$$

$$\underline{E} = \frac{1}{2\mu} \left[\underline{T} - \frac{\lambda}{2\mu + 3\lambda} (\operatorname{tr} \underline{T}) \underline{I} \right] \quad (*)$$

All off-diagonal terms vanish

$$E_{11} = E_{22} = -\frac{1}{2\mu} \frac{\lambda}{2\mu + 3\lambda} \sigma$$

$$E_{33} = \frac{\sigma}{2\mu} \left[\frac{2\mu + 3\lambda - \lambda}{2\mu + 3\lambda} \right] = \sigma \frac{\mu + \lambda}{\mu(2\mu + 3\lambda)} = \frac{\sigma}{E_Y} \text{ where } E_Y = \frac{\mu(2\mu + 3\lambda)}{\mu + \lambda} \text{ Young modulus}$$

$$\text{Then } E_{11} = E_{22} = -\frac{\lambda}{2} \frac{1}{\mu + \lambda} \frac{\mu + \lambda}{\mu(2\mu + 3\lambda)} \sigma = -\nu E_{33} \text{ where } \nu = \frac{\lambda}{2(\mu + \lambda)} \text{ Poisson ratio}$$

$\nu \in (-1, 1/2)$

Can rewrite (*) as

$$\underline{E} = \frac{1}{E_Y} \left[(1 + \nu) \underline{T} - \nu (\operatorname{tr} \underline{T}) \underline{I} \right] \quad (**)$$

Check: $\frac{\nu}{E_Y} = \frac{\lambda}{2\mu(2\mu + 3\lambda)}$ Indeed $\frac{\lambda}{2(\mu + \lambda)} \frac{\mu + \lambda}{\mu(2\mu + 3\lambda)} = \frac{\lambda}{2\mu(2\mu + 3\lambda)} \quad \checkmark$

$\frac{1 + \nu}{E_Y} = \frac{1}{2\mu}$ Indeed $\frac{2\mu + 3\lambda}{2(\mu + \lambda)} \frac{\mu + \lambda}{\mu(2\mu + 3\lambda)} = \frac{1}{2\mu} \quad \checkmark$

From uniaxial extension analysis we have obtained

$$\underline{\underline{E}} = \frac{1}{E_Y} \left[(1+\nu)\underline{\underline{T}} - \nu(\text{tr}\underline{\underline{T}})\underline{\underline{I}} \right] \quad \text{where} \quad E_Y = \frac{\mu(2\mu+3\lambda)}{\mu+\lambda}, \quad \nu = \frac{\lambda}{2(\mu+\lambda)}$$

Insert back to $\underline{\underline{T}} = C \underline{\underline{E}}$

$$\underline{\underline{T}} = \frac{E_Y}{(1+\nu)(1-2\nu)} \left[(1-2\nu)\underline{\underline{E}} + \nu(\text{tr}\underline{\underline{E}})\underline{\underline{I}} \right] \quad (**)$$

Observe that $E_Y \underline{\underline{E}} = (1+\nu)\underline{\underline{T}} - \nu(\text{tr}\underline{\underline{T}})\underline{\underline{I}}$

$$E_Y \text{tr}\underline{\underline{E}} = (1+\nu)\text{tr}\underline{\underline{T}} - 3\nu(\text{tr}\underline{\underline{T}}) = (1-2\nu)(\text{tr}\underline{\underline{T}}) \Rightarrow (\text{tr}\underline{\underline{T}}) = \frac{E_Y}{1-2\nu} (\text{tr}\underline{\underline{E}})$$

$$\text{Then } (1+\nu)\underline{\underline{T}} = E_Y \underline{\underline{E}} + \nu \frac{E_Y}{1-2\nu} (\text{tr}\underline{\underline{E}})\underline{\underline{I}} \Rightarrow \underline{\underline{T}} = \frac{E_Y}{(1+\nu)(1-2\nu)} \left[(1-2\nu)\underline{\underline{E}} + \nu(\text{tr}\underline{\underline{E}})\underline{\underline{I}} \right] \quad \checkmark$$

(**) is the form that allows to write in Voigt notation $[\sigma] = [C][\epsilon]$

$$\text{with } [C] = \frac{E_Y}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & 1-2\nu & 0 & 0 \\ 0 & 0 & 0 & 0 & 1-2\nu & 0 \\ 0 & 0 & 0 & 0 & 0 & 1-2\nu \end{bmatrix}$$

This is easy to generalize to cases with less symmetry than isotropic
(transversely isotropic, orthotropic, ...)

Revisit solution using (**)

$$\underline{T} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sigma \end{bmatrix}, \quad \underline{E} = \frac{1}{E_Y} \left[(1+\nu)\underline{T} - \nu(\text{tr } \underline{T})\underline{I} \right]$$

$$E_{33} = \frac{1}{E_Y} \left[(1+\nu)\sigma - \nu\sigma \right] = \frac{\sigma}{E_Y} = \epsilon \quad \text{extensional strain along the loading direction } x_3$$

$$E_{11} = E_{22} = \frac{1}{E_Y} \left[-\nu\sigma \right] = -\nu\epsilon \quad \text{extensional strain along the unloaded directions } x_1, x_2$$

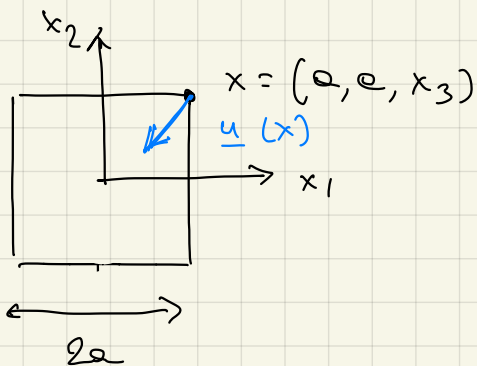
$$\underline{E} = \begin{bmatrix} -\nu\epsilon & 0 & 0 \\ 0 & -\nu\epsilon & 0 \\ 0 & 0 & \epsilon \end{bmatrix} = \text{sym } \nabla \underline{u}$$

This can be integrated (BCs fix the otherwise arbitrary infinitesimal rigid displacement)

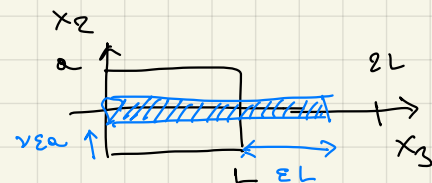
One possible solution is

$$\underline{u}(x_1, x_2, x_3) = -\nu\epsilon x_1 \underline{e}_1 - \nu\epsilon x_2 \underline{e}_2 + \epsilon x_3 \underline{e}_3$$

This solution has
 $\underline{u}(0) = 0$ no transl.
 $\text{skew } \nabla \underline{u} = 0$ no rot.



$$\underline{u}(a, a, x_3) = \begin{bmatrix} -\nu\epsilon a \\ -\nu\epsilon a \\ \epsilon x_3 \end{bmatrix}$$



Remark The solution of linear elasticity is not adequate to study large ϵ .

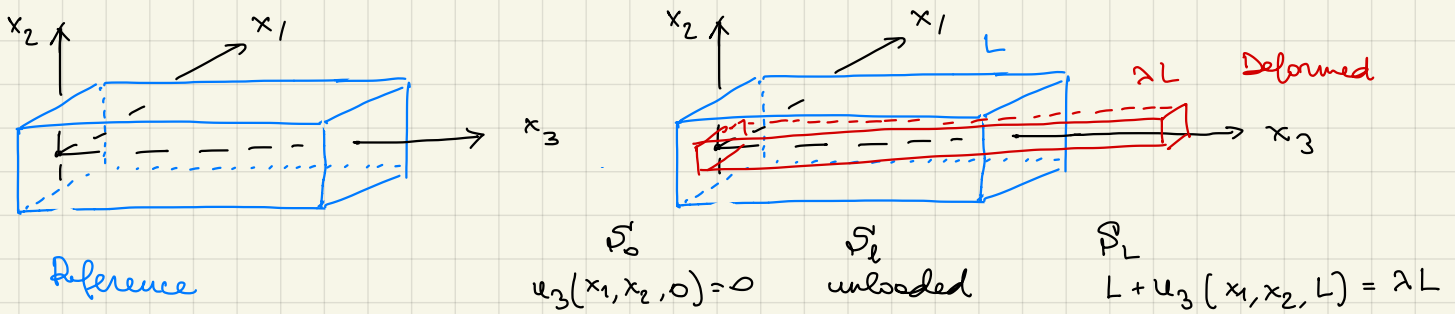
With $\nu > 0$ it gives collapse of cross section and then eversion:

For $\nu\epsilon a = a$ the cross section collapses to a point

For $\nu = \frac{1}{2}$ collapse occurs at $\frac{1}{2}\epsilon = 1 \Rightarrow \epsilon = 2$, a very large strain (!)
 (200% or $L \rightarrow 3L$)

- Need nonlinear elasticity to study these regimes
- The linear model makes no sense for $\epsilon \geq 2$. The linear model is OK for $\epsilon \ll 1$.

COMPLEMENTS : Uniaxial extension (incompressible) in nonlinear elasticity



Incompressibility : $\det F = J = 1$, $\underline{S} = -p J F^{-T} + \frac{\partial W}{\partial F}$

Neo-Hookean : $W(F) = \frac{1}{2} \mu \left[\text{tr}(F^T F) - 3 \right] = \frac{1}{2} \mu \left[F \cdot F - 3 \right]$

$$\Rightarrow \boxed{\underline{S} = -p \frac{\underline{F}^{-T}}{J=1} + \mu \underline{F}}$$

(see complements of uniax. el.)

Try affine solution. $\underline{F} = \begin{bmatrix} \lambda^{1/2} & 0 & 0 \\ 0 & \lambda^{1/2} & 0 \\ 0 & 0 & \lambda \end{bmatrix}$ satisfies $J = 1$, $\underline{F}^{-T} = \begin{bmatrix} \lambda^{1/2} & 0 & 0 \\ 0 & \lambda^{1/2} & 0 \\ 0 & 0 & \lambda^{-1} \end{bmatrix}$

$$\underline{S} = -p \begin{bmatrix} \lambda^{1/2} & & \\ & \lambda^{1/2} & \\ & & \lambda^{-1} \end{bmatrix} + \mu \begin{bmatrix} \lambda^{-1/2} & & \\ & \lambda^{-1/2} & \\ & & \lambda \end{bmatrix} \quad \text{with constant } p \text{ satisfies } \text{div } \underline{S} = 0$$

BCs

$$\underline{S} \underline{e}_1 = 0 : \begin{bmatrix} \underline{S} \\ \underline{S} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = -p \begin{bmatrix} \lambda^{1/2} \\ 0 \\ 0 \end{bmatrix} + \mu \begin{bmatrix} \lambda^{-1/2} \\ 0 \\ 0 \end{bmatrix} = 0 \Rightarrow -p \lambda^{1/2} + \mu \lambda^{-1/2} = 0 \Rightarrow p = \mu \frac{1}{\lambda}$$

$\underline{S} \underline{e}_2 = 0$ gives no new condition

$$\text{Compute force on box } \mathcal{P}_L \text{ with outer normal } \underline{e}_3. \quad \underline{S} = -\mu \begin{bmatrix} \lambda^{1/2} & & \\ & \lambda^{-1/2} & \\ & & \lambda^{-2} \end{bmatrix} + \mu \begin{bmatrix} \lambda^{-1/2} & & \\ & \lambda^{-1/2} & \\ & & \lambda \end{bmatrix} = \mu \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \lambda - \frac{1}{\lambda^2} \end{bmatrix}$$

$$\underline{S} \underline{e}_3 = \mu \begin{bmatrix} 0 \\ 0 \\ \lambda - \frac{1}{\lambda^2} \end{bmatrix}, \quad \text{force } \underline{f} = \int_A \underline{S} \underline{e}_3 dA = \mu A \left(\lambda - \frac{1}{\lambda^2} \right) \underline{e}_3$$

$$\frac{\underline{f}_3}{A} = S_{33}(\lambda)$$

- Check that the cross section collapses to a point only in the limit $\lambda \rightarrow +\infty$.
- Compare here images of uniaxial stretching in linear and nonlinear uniaxial extension of a sample clamped at the ends.

Complements

Constitutive equations : linearly elastic (conservative) material

$$\underline{T} = \underline{C} \underline{E}$$

$\underline{C}$ 4th-order tensor of elastic moduli
(with major symmetry)

Using Voigt's representation

$$[\sigma] = [C][\varepsilon]$$

where $[C]$ is a symmetric 6×6 matrix (21 independent components)

For isotropic materials, two independent parameters

$$\underline{T} = 2\mu \underline{E} + \lambda (\text{tr} \underline{E}) \underline{I}$$

λ, μ Lamé constants ($\mu = G$ shear modulus)

Alternative representations for the isotropic constitutive equation

$$[\sigma] = [C][\varepsilon]$$

(Voigt notation)

$$[C] = \frac{E_Y}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & 1-2\nu & 0 & 0 \\ 0 & 0 & 0 & 0 & 1-2\nu & 0 \\ 0 & 0 & 0 & 0 & 0 & 1-2\nu \end{bmatrix}$$

$$E_Y = \frac{\mu(2\mu + 3\lambda)}{\mu + \lambda}$$

Young modulus

$$\nu = \frac{\lambda}{2(\mu + \lambda)}$$

Poisson ratio

Using $\underline{T} = 2\mu \underline{E} + \lambda (\text{tr } \underline{E}) \underline{I}$ in $\text{div } \underline{T} + \underline{b}_0$ we obtain

$$\mu \Delta \underline{u} + (\lambda + \mu) \nabla (\text{div } \underline{u}) + \underline{b}_0 = 0$$

and note similarity with RHS of Navier Stokes and Stokes.

=

Important theorems for linear elasticity - boundary value problems

. Necessary condition that the traction problem (tractions prescribed all over the boundary) have a solution, is that the force and moment resultants of the external forces are zero

. Uniqueness theorem

The mixed problem of elastostatics (prescribe displacements and tractions on different parts of the boundary) has at most one solution up to an infinitesimal rigid displacement of the body; i.e., any two solutions must have the same stress and strain fields, and the displacement fields of the two solutions may differ at most by an infinitesimal rigid displacement.

. Betti

$\{u_1, b_1, T_1, E_1\}$, $\{u_2, b_2, T_2, E_2\}$ two elastic solutions for the same body (explain)

$$L_{12} = \int_B b_1 \cdot u_2 + \int_{\partial B} T_1 n \cdot u_2 = \int_B \mathbb{C} E_1 \cdot E_2 = \int_B E_1 \cdot \mathbb{C} E_2 = \int_B b_2 \cdot u_1 + \int_{\partial B} T_2 n \cdot u_1 = L_{21}$$

$$\text{e.g. } (2\mu E_1 + \lambda (\text{tr } E_1) \underline{I}) \cdot E_2 = 2\mu E_1 \cdot E_2 + \lambda (\text{tr } E_1) \text{tr } E_2 = (2\mu E_2 + \lambda (\text{tr } E_2) \underline{I}) \cdot E_1$$

Other forms of constitutive equation

$$1) \quad T = 2\mu E + \lambda (\text{tr} E) I$$

$$T = 2\mu \text{dev} E + \kappa (\text{tr} E) I \quad \mu, \kappa \text{ shear and bulk moduli}$$

$$\text{write } T = 2\mu \text{dev} E + 2\mu \frac{1}{3} (\text{tr} E) I + \lambda (\text{tr} E) I$$

$$= 2\mu \text{dev} E + \frac{2\mu + 3\lambda}{3} (\text{tr} E) I \quad \text{and set } \kappa = \frac{2\mu + 3\lambda}{3}$$

The meaning of this constant comes from an experiment of uniform compression / expansion

$$\underline{u}(x) = \varepsilon (x - 0)$$

$$\nabla \underline{u} = \varepsilon \underline{I}, \quad \underline{E} = \varepsilon \underline{I}, \quad \text{tr} \underline{E} = 3\varepsilon \quad \text{volume change}$$

$$\underline{T} = \kappa 3\varepsilon \underline{I}$$

$$\text{average pressure } p = -\frac{1}{3} \text{tr} \underline{T} = -\frac{1}{3} \kappa 3\varepsilon \underline{I} \cdot \underline{I} = -\kappa (3\varepsilon)$$

$\Rightarrow$ κ is the ratio between average pressure and volume change

This form is useful to discover uncoupling in elastic energy

$$\psi = \frac{1}{2} C E \cdot E$$

$$\psi = (2\mu E + \lambda (\text{tr} E) I) \cdot E = 2\mu |E|^2 + \lambda (\text{tr} E)^2$$

$$\psi = (2\mu \text{dev} E + \kappa (\text{tr} E) I) \cdot (\text{dev} E + \frac{1}{3} (\text{tr} E) I) = 2\mu |\text{dev} E|^2 + \kappa (\text{tr} E)^2$$

$$2) \quad T = 2\mu \underline{E} + \lambda (\text{tr} \underline{E}) \underline{I}$$

find inverse relation

$$\underline{E} = \frac{1}{2\mu} \left[\underline{T} - \frac{\lambda}{2\mu + 3\lambda} (\text{tr} \underline{T}) \underline{I} \right]$$

$$\text{Observe that } \text{tr} \underline{T} = 2\mu \text{tr} \underline{E} + \lambda (\text{tr} \underline{E}) (\text{tr} \underline{I}) = (2\mu + 3\lambda) \text{tr} \underline{E}$$

$$\text{then } \frac{1}{2\mu} \left(\underline{T} - \frac{\lambda}{2\mu + 3\lambda} (\text{tr} \underline{T}) \underline{I} \right) = \underline{E}$$

3) From uniaxial extension (see later) we obtain

$$\underline{E} = \frac{1}{E_Y} \left[(1+\nu) \underline{T} - \nu (\text{tr} \underline{T}) \underline{I} \right] \quad \text{where } E_Y = \frac{\mu(2\mu + 3\lambda)}{\mu + \lambda}, \quad \nu = \frac{\lambda}{2(\mu + \lambda)}$$

Find inverse relation

$$\underline{T} = \frac{E_Y}{(1+\nu)(1-2\nu)} \left[(1-2\nu) \underline{E} + \nu (\text{tr} \underline{E}) \underline{I} \right] \quad (*)$$

$$\text{Observe that } E_Y \underline{E} = (1+\nu) \underline{T} - \nu (\text{tr} \underline{T}) \underline{I}$$

$$E_Y \text{tr} \underline{E} = (1+\nu) \text{tr} \underline{T} - 3\nu (\text{tr} \underline{T}) = (1-2\nu) (\text{tr} \underline{T}) \Rightarrow (\text{tr} \underline{T}) = \frac{E_Y}{1-2\nu} (\text{tr} \underline{E})$$

$$\text{Then } (1+\nu) \underline{T} = E_Y \underline{E} + \nu \frac{E_Y}{1-2\nu} (\text{tr} \underline{E}) \underline{I} \Rightarrow \underline{T} = \frac{E_Y}{(1+\nu)(1-2\nu)} \left[(1-2\nu) \underline{E} + \nu (\text{tr} \underline{E}) \underline{I} \right] \quad \checkmark$$

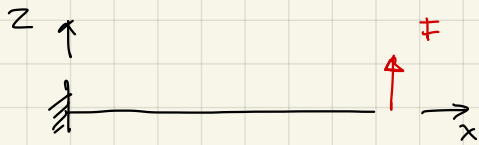
(*) is the form that allows to write in Voigt notation $[\sigma] = [C][\epsilon]$

$$\text{with } [C] = \frac{E_Y}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & 1-2\nu & 0 & 0 \\ 0 & 0 & 0 & 0 & 1-2\nu & 0 \\ 0 & 0 & 0 & 0 & 0 & 1-2\nu \end{bmatrix}$$

Lecture 8 : Linear Elasticity in COMSOL

14/4/26

Benchmark for bending from linear theory of rods / beams



$$EI w^{IV} = f_z = 0 \Rightarrow w(x) = a + bx + cx^2 + dx^3$$

$$w(0) = a = 0$$

$$w'(0) = b = 0$$

$$w(x) = cx^2 + dx^3$$

$$w'(x) = 2cx + 3dx^2$$

$$w''(x) = 2c + 6dx$$

$$w'''(x) = 6d$$

$$w''(l) = 2c + 6dl = 0 \Rightarrow c = -3dl$$

$$-EI w'''(L) = F \Rightarrow d = -\frac{1}{6} \frac{F}{EI}$$

$$\Rightarrow c = \frac{1}{2} \frac{Fl}{EI}$$

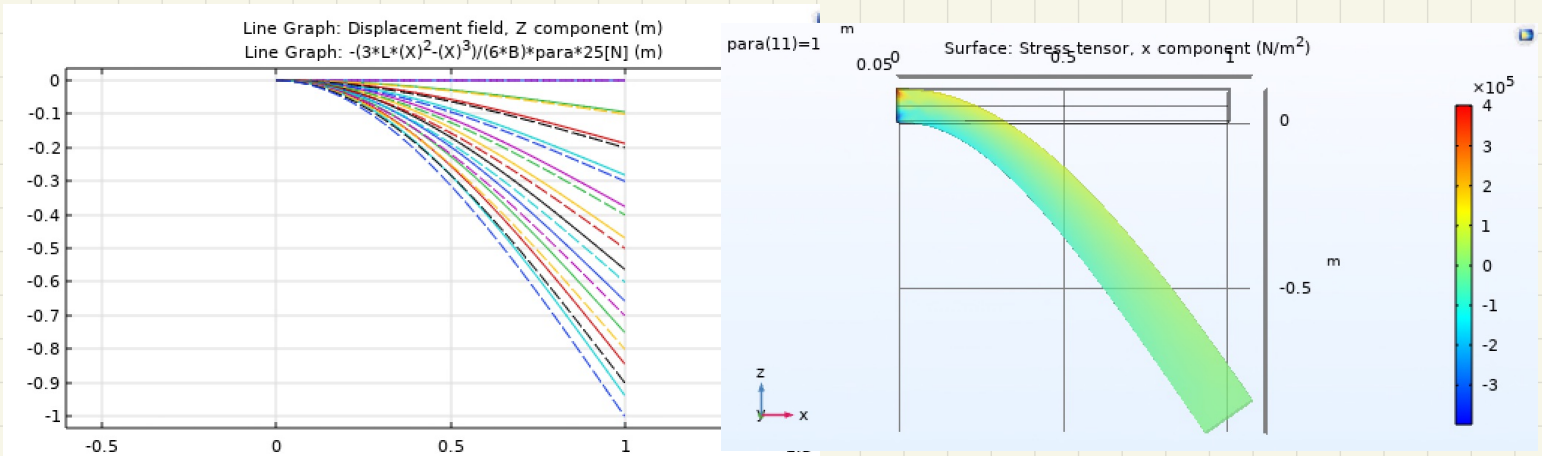
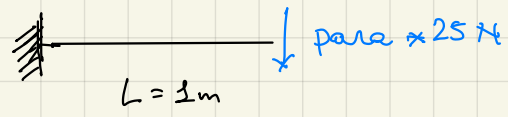
$$w(x) = \frac{1}{2} \frac{Fl}{EI} x^2 - \frac{1}{6} \frac{F}{EI} x^3 = \frac{F}{6EI} (3lx^2 - x^3)$$

$$\bullet \quad w(l) = \frac{F}{6EI} (3l^3 - l^3) = \frac{Fl^3}{3EI}$$

$$\bullet \quad w(l) = l \quad \text{for} \quad F = \frac{3EI}{l^2}$$

Comparison of cantilever bending : linear and nonlinear elasticity

Linear 3D vs linear analytic 1D



Nonlinear 3D vs linear analytic 1D

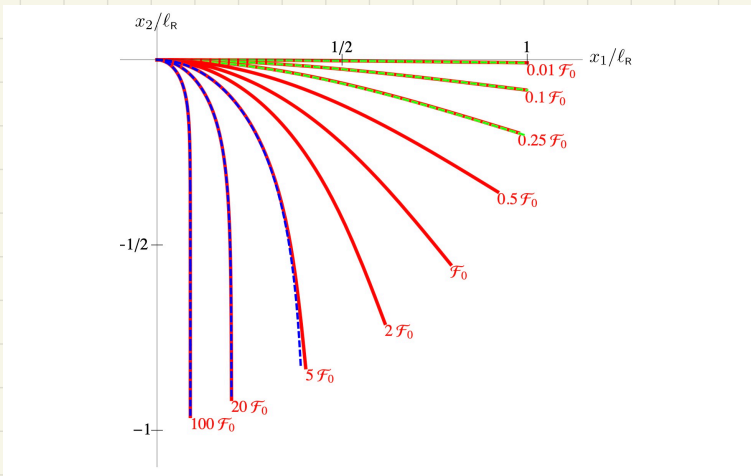
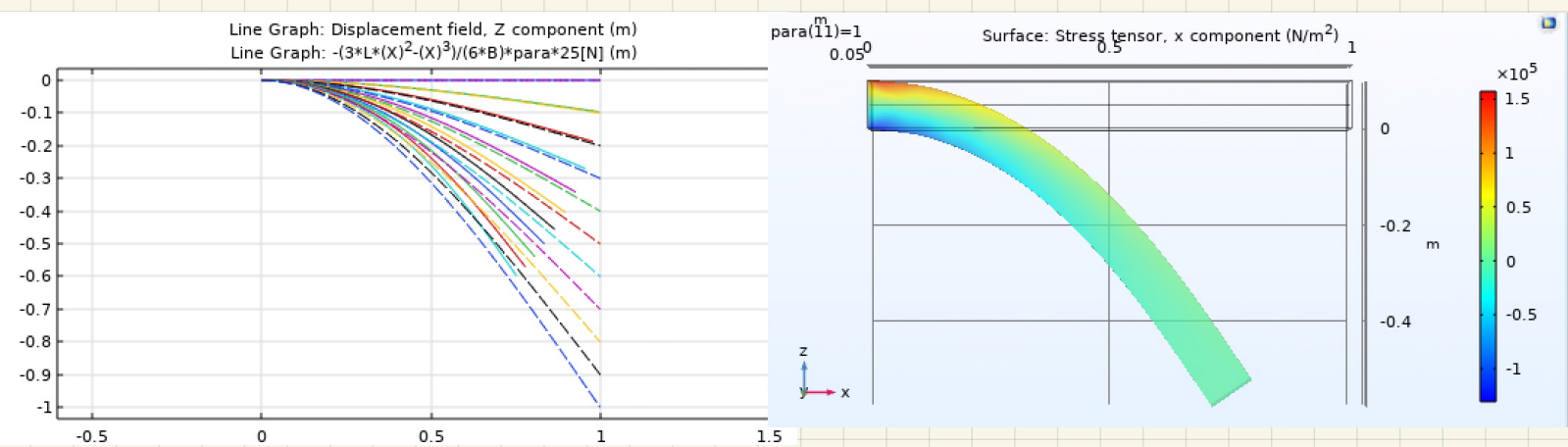
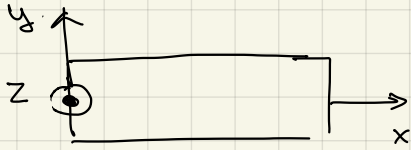
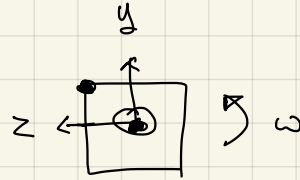


FIGURE 4.21 – Comparaison des différents modèles dans le cas d’une force verticale ($\omega = -\pi/2$) en l’absence de pesanteur. En rouge : la configuration d’équilibre donnée par le modèle de tige inextensible; en pointillé vert : celle donnée par le modèle linéarisé de flexion; en pointillé bleu : celle donnée par le modèle de fil inextensible avec prise en compte de la couche limite à l’encastrement.

Displacement field for torsion



$$\underline{\underline{\omega}} = \omega \underline{e}_1$$



$$\begin{aligned}\underline{v}(l, y, z) &= \omega \underline{e}_1 \wedge (y \underline{e}_2 + z \underline{e}_3) \\ &= \omega (y \underline{e}_3 - z \underline{e}_2)\end{aligned}$$

$$\underline{u} = \underline{v} dt = \underbrace{\omega dt}_{\Theta} (y \underline{e}_3 - z \underline{e}_2)$$

$$u_2 = -\Theta z$$

$$u_3 = \Theta y$$